University of Sunderland

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MODULE CODE: CET138

MODULE TITLE: Full Stack Development

MODULE LEADER: David Grey

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ASSIGNMENT: 1

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- **Unauthorised use of artificial intelligence** – where you use artificial intelligence tools to generate your assignment instead of completing it yourself and/or where you have not been given permission to use artificial intelligence tools by your module leader. Please complete the following declaration around you use of artificial intelligence tools in your assignment.

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• I have used artificial intelligence tools to generate an idea for my assignment:	**NO**
• I have used artificial intelligence tools to write my assignment for me:	**NO**
• I have used artificial intelligence tools to brainstorm ideas for my assignment:	**YES**
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DECLARATION

- I understand that by submitting this piece of work I am declaring it to be my own work and in compliance with the university regulations on Academic Integrity.
- I confirm that I have done this work myself without external support or inappropriate use of resources.
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- I confirm that this piece of work has not been submitted for any other assignment at this or another institution prior to this point in time.
- I can confirm that all sources of information, including quotations, have been acknowledged by citing the source in the text, along with producing a full list of the sources used at the end of the assignment.
- I understand that academic misconduct is an offence and can result in formal disciplinary proceedings.
- I understand that by submitting this assignment, I declare myself fit to be able to complete the assignment and I accept the outcome of the assessment as valid and appropriate.

Live Demo Link

https://ydvbishal4869-sketch.github.io/demo/js-demo.html

JavaScript

If HTML is the **skeleton** that holds everything up and CSS is the **skin and clothing** that makes it look good, JavaScript is the **brain, muscles, and nervous system**. Without it, a website is just a mannequin—nice to look at, but completely lifeless. JavaScript is the programming language that gives a webpage a pulse, making it interactive, adaptive, and alive.

What Does JavaScript Actually Do?

Think of HTML and CSS as static. They just sit there looking pretty. JavaScript, on the other hand, is all about **action**. It constantly watches how a user interacts with the page and changes things on the fly—no frustrating page refreshes required.

Here is what that looks like in the real world:

- **It Reacts to You:** It notices when you click a button, type in a box, or scroll down a page, and it triggers an immediate response.
- **It Thinks Ahead (Form Checking):** When you type in a password, JavaScript is the reason you instantly see *"Passwords don't match"* or *"Invalid email"* before you even hit submit.
- **It Changes the Scenery Dynamically:** It can pop up a notification, swap an image, or add new text to your screen instantly based on what you're doing.
- **It Brings Movement:** From smooth image sliders and drop-down menus to live countdown timers, JavaScript handles the website's behavior and flow.

Anatomy of a JavaScript Action: How It Thinks

- JavaScript essentially operates on a simple, three-step human logic: **Select, Listen, and Act**.
- Behind the scenes, it hooks into the HTML structure (which developers call the **DOM**, or Document Object Model) and follows a basic rule:
- **"Find** *this* specific thing on the page, **wait** for the user to do *that* to it, and then **execute** this exact reaction."

Here is a basic example of how it looks:

JavaScript

```
// 1. Find the button on the page
const registerButton = document.querySelector('.submit-btn');

// 2. Listen for when someone clicks it
registerButton.addEventListener('click', function() {
// 3. Do something cool!     alert('Thank you for
clicking the register button!');
```

});

How to Add JavaScript to Your Registration Form

Let's make your form smart! Right now, if a user clicks "Register", nothing happens. Let's add a quick validation feature so that if they try to submit the form without filling out their full name, an alert pops up.

Step 1: Open your text editor Open

a new blank window.

Step 2: Paste this JavaScript code

```
JavaScript
// Wait until the webpage is completely loaded document.addEventListener('DOMContentLoaded',
() => {

   // Find our form using its tag
   const form = document.querySelector('form');

   // Listen for the "submit" event when someone clicks Register
form.addEventListener('submit', (event) => {

      // Grab the value inside the Full Name input field
      const fullName = document.getElementById('fullname').value.trim();

      // Check if the name field is completely empty        if
(fullName === '') {           event.preventDefault(); // Stop the
form from submitting!           alert('Please enter your Full Name
before registering.');
      } else {
         alert('Form submitted successfully, ' + fullName + '!');
      }
   });
});
```

Step 3: Save the File

1. Go to **File > Save As...**
2. Save it in the **same folder** (RegistrationForm) alongside your HTML and CSS.
3. Name it **script.js** (change the save type dropdown to **All Files (.)** and use quotation marks like "script.js" to avoid the .txt issue).

Step 4: Link it in your index.html

To make your HTML file read your JavaScript, open your index.html file and add this single line right before the closing </body> tag at the very bottom:

HTML
```
   <script src="script.js"></script>

</body>
</html>
```

Simple calculator Demo code

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Simple Calculator</title>


<style>

   *{

       margin:0;        padding:0;

box-sizing:border-box;        font-

family:Arial, sans-serif;

   }


   body{        height:100vh;

display:flex;        justify-

content:center;        align-

items:center;

       background:linear-gradient(135deg,#667eea,#764ba2);
```

```
    }

    .calculator{
        background:white;       padding:30px;
border-radius:15px;        box-shadow:0 10px
25px rgba(0,0,0,0.3);        text-align:center;
width:320px;
    }

    h2{        margin-
bottom:20px;
        color:#333;
    }

    input{        width:100%;
padding:12px;
margin:10px 0;
border:1px solid #ccc;
border-radius:8px;
font-size:16px;
    }

    .buttons{
display:grid;
        grid-template-columns:repeat(4,1fr);
gap:10px;        margin-top:15px;
    }
```

```
   button{
padding:12px;
border:none;        border-
radius:8px;
background:#667eea;
       color:white;        font-
size:18px;        cursor:pointer;
transition:0.3s;
   }

   button:hover{
background:#764ba2;
transform:scale(1.05);
   }

   #result{        margin-
top:20px;        font-
size:20px;        font-
weight:bold;
color:#333;
   }
</style>
</head>

<body>

<div class="calculator">
   <h2>Simple Calculator</h2>     <input type="number" id="num1" placeholder="Enter First Number">
```

```
    <input type="number" id="num2" placeholder="Enter Second Number">

    <div class="buttons">
        <button onclick="calculate('+')">+</button>
        <button onclick="calculate('-')">−</button>
        <button onclick="calculate('*')">×</button>
        <button onclick="calculate('/')">÷</button>
    </div>

    <div id="result">Result: 0</div>
</div>

<script> function
calculate(operator){

    let num1 = parseFloat(document.getElementById("num1").value);
let num2 = parseFloat(document.getElementById("num2").value);

    if(isNaN(num1) || isNaN(num2)){
document.getElementById("result").innerHTML =
        "Please enter valid numbers";
        return;
    }

    let result;

    switch(operator){

        case '+':
```

```
            result = num1 + num2;

            break;

         case '-
':

            result = num1 - num2;

            break;


        case '*':

            result = num1 * num2;

            break;

         case
'/':

            if(num2 === 0){                result =
"Cannot divide by zero";

            }else{                result =
num1 / num2;

            }

            break;

    }


    document.getElementById("result").innerHTML =

    "Result: " + result;

}

</script>


</body>

</html>
```